

ADITYA RAMESH

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EDUCATION

University of Illinois at Urbana-Champaign

May 2026

Bachelor of Science in Statistics and Computer Science

Urbana-Champaign, IL

Relevant Courses: Distributed Systems, Cloud Data Management, Compilers, Data Structures, Algorithms, Computer Systems, Database Systems, Applied ML, Statistical Modeling

TECHNICAL SKILLS

Languages: Python, C/C++, Go, SQL, Java, Bash, R

Libraries & Frameworks: PyTorch, HuggingFace, LangChain, NumPy, Pandas, Scikit-learn, Keras, Flask

Tools & Platforms: GCP, AWS, Docker, Redis, Git

PROFESSIONAL EXPERIENCE

Synchrony Financial

Jan 2026 – May 2026

Software Engineering Intern

- Engineered an **AWS EMR** and **OpenLineage** POC to migrate TB-scale historical (13-years) data from legacy **Hadoop**.
- Automated data discovery via Marquez on **EC2**, reducing manual credit profile auditing from 2 months to 3 days.
- Added validation and failure handling for malformed records and jobs to ensure idempotent runs and consistency.

Motorola Solutions

Feb 2025 – Dec 2025

Software Engineering Intern

- Engineered real-time, low-bitrate (4.4 kbps) AI Vocoders providing field-deployable wideband audio to first responders.
- Parallelized PESQ/POLQA calculations via **Bash** scripts, slashing internal testing runtime from ~48 hours to 5 hours.
- Benchmarked STT services (**Azure**, **AWS**, **GCP**) using WER/Intent metrics, informing translation architecture decisions.
- Deployed **PyTorch Mobile** models onto embedded Android systems, enabling field-ready inference for APX NEXT radios.

FOCAL Lab at UIUC

Aug 2024 – Dec 2024

AI Undergraduate Researcher

- Extended UIUC's QuaCer-B framework to detect intersectional biases using **LlamaGuard** for **LLM-as-a-judge** evaluation.
- Architected a scalable **Python** evaluation pipeline to concurrently test multiple LLMs against 1600 unique vectors.
- Proved feasibility of scalable, multi-attribute LLM bias detection, winning Best Presentation at UIUC URSA Symposium.

ATLAS at UIUC

May 2024 – Dec 2024

Machine Learning Intern

- Implemented a **computer-vision** enabled trash/recycling classifier for UIUC, enhancing waste-sorting efficiency by 43%.
- Parallelized preprocessing of 17,000 images via **pandas/NumPy**, eliminating YOLOv9 **CNN** training-pipeline bottlenecks.

PROJECTS

Bytecode Compiler and Runtime Engine

- Constructed a register-based VM in **C++** to compile and execute custom assembly via an optimized bytecode runtime.
- Implemented a compiler frontend with zero-copy lexing, **ASTs**, monadic error-handling, and IR constant-folding.
- Engineered a 64MB **memory arena** and lock-free **SPSC ring buffers**, eliminating false-sharing, improving cache locality.

ML-Based Packet Anomaly Detector

- Developed a packet-sniffer in **C++** using **libtins**, capable of capturing, parsing, and serializing packets in 0.05 ms.
- Engineered a real-time processing pipeline using **Redis** circular buffers and the **GeoLite2 API** to enrich traffic metadata.
- Trained an anomaly-detection **Isolation Forest** via **pandas** and **scikit-learn** for suspicious network-activity flagging.

GCP Serverless Elo Ranking Platform

- Architected a serverless anime-ranking platform for concurrent users via **Flask**, **Cloud Run**, **Firestore**, & **Firebase**.
- Integrated dual-layer ttl-caching and atomic batch updates, minimizing database reads during 30,000+ daily queries.
- Built a **Python GraphQL** pipeline via AniList to clean and consolidate 200 franchises for personal & global leaderboards.

Confluence Knowledge-Retrieval Assistant

- Created a knowledge-retrieval assistant that grounds LLM responses in internal **Confluence** documentation using **RAG**.
- Implemented semantic and lexical retrieval pipelines with **HuggingFace** embeddings and **Pinecone** vector search.
- Deployed a local **Llama 3.1** runtime through **Ollama** with **Redis** for conversation history persistence across users.